Technical Case Study: Decision Analytics Reconstruction

Architecture, methodology, and implementation decisions for a three-module national-scale decision analytics system

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System Overview

This document describes the technical architecture and implementation decisions of a three-module decision analytics system built as a practitioner reconstruction. The system processed 4,260,816 entities across 18 geographic units, allocated a constrained budget across 11 channel types over 18 weeks, and produced a calibrated probabilistic forecast of a verifiable binary outcome.

Outcome: Program achieved objective by +3.70 pp (Series A) and +3.88 pp (Series B). Verified by TSJE 2018.

Module A: Population Modeling and Segmentation

Architecture

```
 generation.yaml          → generator.py          → population_raw.parquet
calibration_anchors.yaml → raw_injector.py       → population_raw_flawed.parquet
model_params.yaml        → cleaner.py            → population_master_clean.parquet
                         → features/demographic  → (engineered features)
                         → features/behavioral   → (interaction terms)
                         → features/reachability → (channel access scores)
                         → models/segmentation   → segment_labels.parquet
                         → models/propensity     → participation_propensity.parquet
                         → evaluation/validator  → qa_report.md
                         → app/streamlit_dashboard
```

Synthetic data generation

The raw population is generated from verified demographic anchors. Key design decisions:

IPF/raking. Simple `rng.choice()` with probability weights produces marginal distributions that can drift from anchors due to interaction effects (e.g., rural × language). Two raking passes are applied: (1) binary raking of `rural_flag` to the verified 38.3% rural share via random bit-flipping, (2) categorical raking of `language_census_bucket` to DGEEC bilingualism marginals. This is a simplified IPF; full iterative proportional fitting is deferred to the cleaner.

Department weights. Electoral-roll population shares (TSJE 2018) normalized to sum exactly to 1.0. Eighteen departments including Asunción (treated as a separate administrative unit). Weight for Central ≈ 32.6%, consistent with Central being the most populous department.

Flaw injection. Thirteen realistic data quality problems are injected into the raw layer by `raw_injector.py`, calibrated to observed rates in real administrative data collection:

Flaw type	Rate	Mechanism
Cédula format error	3.5%	Missing zero-pad on 7-digit numbers
Duplicate registration	1.2%	Cross-office re-entry simulation
Department typo	2.5%	"Cordilera", "Caaguazu" variants
Municipality null	8.0%	Blank field from field collection
Date format swap	1.8%	MM/DD/YYYY from 2 offices
Encoding error	4.0%	Windows-1252 → UTF-8 garble
Phone format variant	4.5%	Three observed formats
Gender variant	3.0%	"Masculino", "1", "2"
Age range error	0.8%	Transposed DOB producing age <18 or >120
Schema drift	1.5%	Column naming inconsistency
Sentiment scale	20.0%	Inconsistent 1–5 scale across offices
District null	25.0%	Neighborhood name not collected
Rural flag absent	100%	Always derived; never in raw

The 14-step cleaning pipeline in `cleaner.py` addresses each flaw type in documented sequence (see `reports/transformation_log.md`).

Segmentation model

Pre-pass: DBSCAN identifies structural noise entities (those with unusual feature combinations that do not belong to any behaviorally coherent cluster). Noise rate target: < 1%. Epsilon and min_samples tuned on a 10K sample; validated on full population.

K-Means (k=6) applied to the non-noise population on a 14-feature matrix:

```
 age_bin_encoded, gender_encoded, youth_flag, senior_flag,
language_jopara_encoded, nbi_stress_prior_scaled,
structural_dependency_encoded, preference_proxy_encoded,
reachability_digital, reachability_broadcast_tv,
reachability_broadcast_radio, metro_flag, chaco_flag,
rural_offline_compound
```

Acceptance criteria: mean silhouette > 0.35, bootstrap ARI > 0.80 across 100 seeds.

The six resulting segments have interpretable behavioral profiles (see model card). The segment names are operationally meaningful labels assigned post-hoc, not clustering inputs.

Propensity model

Model: Logistic regression with L2 regularization, trained on synthetic population with known participation outcomes (derived from calibration anchors + noise).

Calibration: Platt scaling post-hoc to linearize the sigmoid. Reliability diagram checked per decile; max deviation target < 3 pp.

Department raking: Post-calibration rake multiplier per department, computed from the ratio of verified departmental participation rate (calibration_anchors.yaml) to model-predicted mean. Applied as a multiplicative correction to raw logit before sigmoid.

Acceptance criteria:

Metric	Target	Rationale
AUC-ROC	> 0.70	Better than stratified baseline
Brier score	< 0.22	10 pp improvement over naive Brier (0.238 at p=0.6125)
Reliability diagram	< 3 pp per decile	Per scope §7.3
National mean propensity	61.25% ± 0.1 pp	Matches verified TSJE anchor

SHAP analysis: LinearExplainer on 10K-entity sample. Feature importance used to validate that the model is learning from behavioral signals, not from artifacts of the generation process.

Module B: Resource Allocation Engine

Problem formulation

Mixed-integer linear program over the decision space: `x[d, c, w]` = budget allocated to department `d`, channel `c`, week `w`.

Objective: Maximize total persuasion-adjusted contacts:

```
max Σ_{d,c,w}  x[d,c,w] / unit_cost[d,c]
             × persuasion_weight[segment_affinity(d,c)]
             × (1 - reach_utilisation[d,c,w] / reach_cap[d,c])
```

Constraints: - `Σ_c,w x[d,c,w] ≤ budget_envelope[d]` (department budget cap) - `Σ_w x[d,c,w] / unit_cost[d,c] ≤ reach_cap[d,c]` (population coverage ceiling) - `x[d,c,w] ≥ 0` (non-negativity) - `Σ_d x[d,c,w] / unit_cost[d,c] ≥ coverage_min × population[d]` for ≥80% of municipalities - FX constraint: `TC_ref_used ∈ [5,500, 5,700] PYG/USD` (BCP Q1 2018 corridor)

Solver: PuLP with CBC backend. OPTIMAL status required; INFEASIBLE halts execution and triggers systematic debugging before any constraint relaxation.

FX modeling

The Guaraní/USD rate is modeled as a weekly time series with:

- Reference rate from BCP daily TC_Ref (Q1 2018: 5,500–5,700 PYG/USD)
- Retail spread: ~+50 PYG/USD above reference
- Scenario bands: baseline (5,600), strengthening (5,500), weakening (5,700)

All budget figures reported in both USD and PYG to separate operational decisions from FX exposure.

API layer

FastAPI endpoint for re-optimization with updated inputs (new segment scores, updated FX rate). P95 latency target: ≤ 2 seconds under 10 concurrent requests.

Module C: Probabilistic Forecasting

Model architecture

Bayesian hierarchical tracking model built in PyMC. The generative model:

```
 preference_margin[t] ~ Normal(mu[t], sigma_obs)
mu[t] = alpha + Σ_j beta[j] * X[t,j] + house_effect[pollster]
alpha ~ Normal(prior_mean, prior_sd)   # calibrated to pre-campaign anchor
house_effect[k] ~ Normal(0, sigma_house)
sigma_house ~ HalfNormal(5.0)          # house effect SD < 5 pp target
sigma_obs ~ HalfNormal(10.0)
```

Sampling diagnostics (acceptance criteria)

Diagnostic	Target	Rationale
R-hat (all vars)	< 1.01	ArviZ/PyMC standard
ESS_bulk	> 400	Reliable posterior summary
ESS_tail	> 400	Reliable tail estimates
Divergences (post-tuning)	0	No geometry pathologies

Any violation blocks delivery. Reparameterization requires systematic debugging (4-phase protocol) before any prior change.

House effect correction

Four polling sources showed structural biases:

Pollster	Estimated house effect	Direction
ATI/Snead	−5.1 pp	Understates Series A
ICA	+3.8 pp	Overstates Series A
CAPLI	−0.4 pp	Near-unbiased (lowest transparency rating)
OEA/EU	−1.1 pp	Mild Series A understatement

The model treats house effects as unknown parameters with a shrinkage prior, estimated jointly with the preference trajectory.

Scenario engine (Monte Carlo)

10,000 Monte Carlo draws simulate shock scenarios:

- **Baseline tracker:** Moderate shocks (scale ~1.0×)
- **Moderate tracker:** Medium shocks (scale ~1.5×)
- **Extreme tracker:** Large shocks (scale ~1.8–2.4×)

Shock sources: late-breaking adverse events, polling infrastructure failures, turnout depression in Youth Volatile segment.

Known pipeline issue (flagged): `alloc_mean_persuasion_contacts` is currently unlinked from Module B solver outputs, meaning the MC model cannot yet quantify how budget reallocation affects win probability. This is the priority cross-module integration fix.

Quarto deliverable

Module C outputs a rendered Quarto document (`module_c_forecasting_scenarios/portfolio/quarto/post_mortem.qmd`) containing the full post-mortem analysis. `quarto render` exit 0 is a required delivery gate.

Engineering Standards

Test-driven development

Every `src/` change follows the TDD iron law: failing test first → minimal implementation → green. The test suite is CI-gated with an 80% coverage floor.

```
 138 tests — Module A (pytest, includes pandera schema contracts)
153 tests — EDA validation (pytest)
  0 failures across both suites
```

Schema contract enforcement

Column names are never bare strings in `src/`. All column names are `Final` typed constants in `schema.py`, imported explicitly. The raw layer uses `_raw` suffixed constants where the clean layer uses clean names (e.g., `ENC_SOURCE_RAW` in generator output, `ENC_SOURCE` in cleaner output). Schema contracts are versioned YAML files in `schema_contracts/`.

Reproducibility

- All random operations use an explicitly seeded `numpy.random.Generator` from `seeds.py`
- `RANDOM_SEED` environment variable sets the global seed; defaults to `20180422` (outcome event date)
- All outputs are version-pinned via run manifests (JSON sidecar files)
- MLflow experiment tracking for model runs

Data quality gates (QA)

Thirteen quantitative acceptance gates are checked by `evaluation/validator.py` before any downstream module receives Module A outputs. A `QAGateFailure` exception halts the pipeline and requires documented resolution before the gate is reopened.

Repository Structure

```
decision-analytics-reconstruction/
├── module_a_population_segmentation/    # [FLAGSHIP] Core pipeline
│   ├── src/population_segmentation/
│   │   ├── data/          # generator, cleaner, raw_injector
│   │   ├── features/      # demographic, behavioral, reachability
│   │   ├── models/        # segmentation, propensity
│   │   ├── evaluation/    # validator, calibration_metrics, clustering_metrics
│   │   ├── visualization/ # segment_profiles, calibration_curves
│   │   └── pipeline/      # export
│   ├── config/            # generation.yaml, calibration_anchors.yaml, model_params.yaml
│   ├── tests/             # 116 tests, TDD-compliant
│   ├── reports/           # model cards, QA reports, transformation log
│   └── app/               # Streamlit dashboard
├── module_b_resource_allocation/        # LP solver, FX layer, FastAPI
├── module_c_forecasting_scenarios/      # PyMC models, scenario engine, Quarto
├── schema_contracts/                    # Versioned YAML column contracts
├── appendix/                            # Verified calibration anchors
└── reports/                             # Case studies, EDA, decision log
```

Skills Demonstrated

Skill	Where demonstrated
Statistical modeling (Bayesian)	Module C: PyMC hierarchical model, MCMC diagnostics
Machine learning (classical)	Module A: Logistic + Platt, DBSCAN + KMeans, SHAP
Optimization (LP)	Module B: PuLP/CVXPY constrained solver
Data engineering	Module A: 14-step cleaning pipeline, IPF raking, schema contracts
Software engineering	TDD, CI/CD, schema constants, reproducible seeds, Docker
Technical writing	Model cards, transformation log, QA reports, Quarto
Uncertainty quantification	Bayesian HDI, Monte Carlo scenario engine, reliability diagrams
Domain expertise	Demographic calibration, FX modeling, media channel optimization